



(Don't Write anything in this Area)

Q.2) (a) Analyse the role of genetics in understanding human diversity. How do modern genetic studies challenge traditional notions of race? (20 marks)

Genetics has become integral part of ~~the~~ Physical anthropology concerned with evolution & variation of human beings across space & time, transforming the discipline into Biological Anthropology.

Role of Genetics in understanding human diversity

* Genetic Markers

They refer to ~~the presence or absence of~~ certain identified genes whose presence or absence is indicative of genetic differences between the given populations.

For Ex, Transferrin can be one such genetic marker. Transferrin is blood protein that carries haemoglobin and the

their concentration can be altered to divide the population based on this diversity

ABO Blood group

It divides the entire population & notes what makes certain population diverse in having A type, B type, O type, etc based on its advantageous role played by such blood groups.

Cytogenetics

It helps in picturing the chromosomes present within human cell. The resultant Karyotype, helps in identifying human diversity in terms of diseased population or normal, etc

Population genetics

It studies the entire genome of given population along with all the allelic differences

possible. It strives to give genetic maps
of human diversity

Modern Genetics challenging traditional notion
of race

'Race' is a social construct. Although
emerged as misapplication of Mendelian genetics
during 1900s, the science of genetics
itself has given the solution in
debunking it scientifically.

Human Genome Project

It was started in 2003 by
National Institute of Health, USA with
an objective to decode entire human
DNA sequence.

It has revealed that
99.99% of human beings are similar in
genetic composition. The 0.001% is not

large enough to create racial divisions.

Clinical model of gene genetics

It says that no single human population has ^{all} exclusive gene & no single gene is exclusively present in particular population. The interbreeding at ^{so called} ~~these~~ ^{these} 'boundaries' has led to gradual increase or decrease of particular genetic frequency.

Genetic flow

The intermixing of races has ^{been} found to be more beneficial than harm in any society.

It leads to increase in genetic pool of a given population offering them with multitude of choices to face environmental changes.

Therefore, As UNESCO made its declaration in 1990; that there is no such thing as racial superiority & all humans are equal.